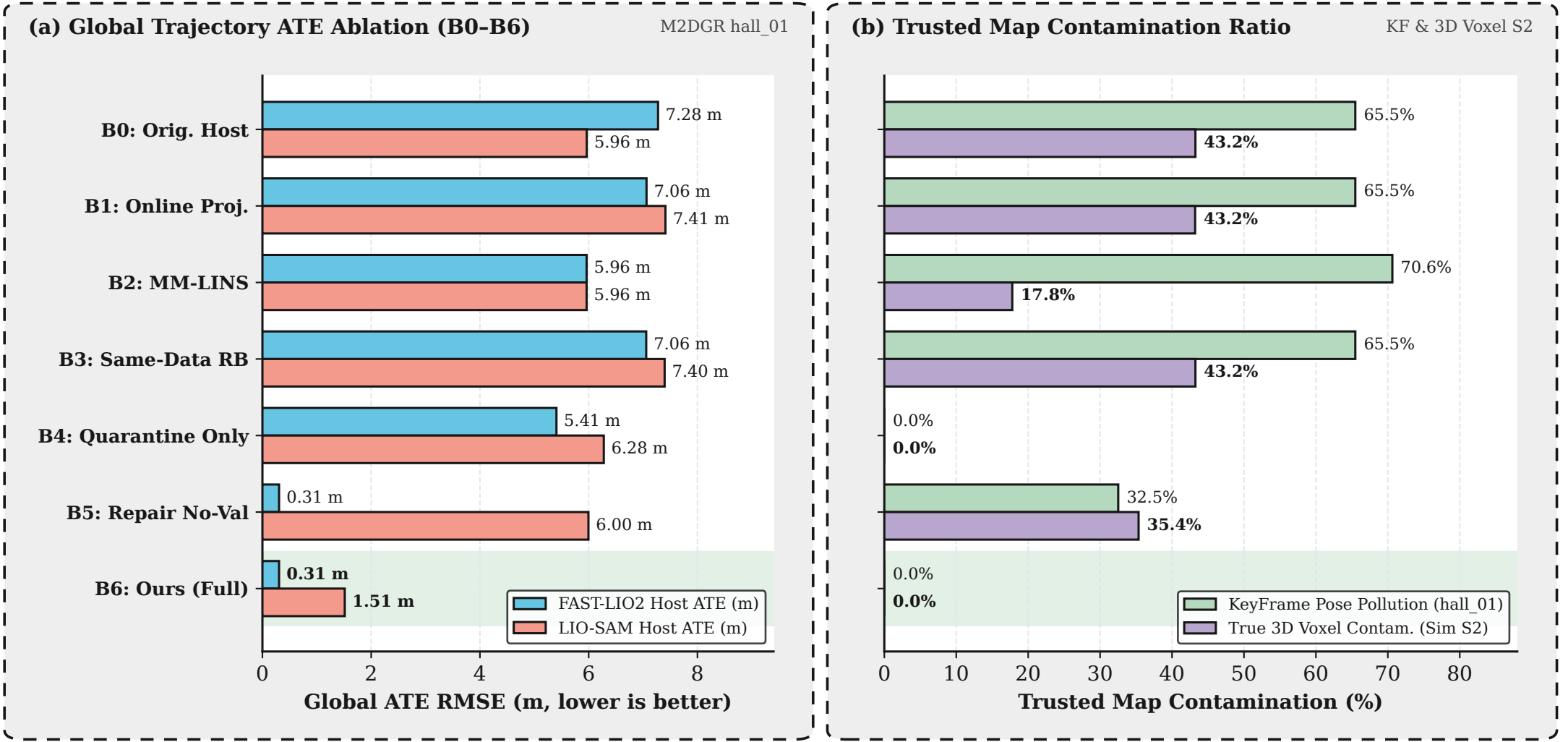
(a) Global Trajectory ATE Ablation (B0–B6)
M2DGR hall_01
B0: Orig. Host
B1: Online Proj.
B2: MM-LINS
B3: Same-Data RB
B4: Quarantine Only
B5: Repair No-Val
B6: Ours (Full)
7.28 m
5.96 m
7.06 m
7.41 m
5.96 m
5.96 m
7.06 m
7.40 m
5.41 m
6.28 m
0.31 m
6.00 m
0.31 m
1.51 m
FAST-LIO2 Host ATE (m)
LIO-SAM Host ATE (m)
0
2
4
6
8
Global ATE RMSE (m, lower is better)
(b) Trusted Map Contamination Ratio
KF & 3D Voxel S2
65.5%
43.2%
65.5%
43.2%
70.6%
17.8%
65.5%
43.2%
0.0%
0.0%
32.5%
35.4%
0.0%
0.0%
KeyFrame Pose Pollution (hall_01)
True 3D Voxel Contam. (Sim S2)
0
10
20
30
40
50
60
70
80
Trusted Map Contamination (%)